Development of a Multi-Stage Analog Signal Conditioning Circuit Using BJT and RC Filters

Module Name: Analog Electronics – IE2034

Group Members:

IT24103620 -Viyashan.K

IT24103482 - Lahinth.M

Contents:

Introduction

The signals captured by the antenna in simple AM radio systems are frequently weak and combined with a lot of background noise. These weak signals need to go through several filtering and amplification processes to be converted into clear audio. The design and construction of a three-stage analog signal conditioning circuit using RC filter and Bipolar Junction Transistors (BJTs) is the main goal of this project.

Objective

1. To design a first order high-pass RC filter with a cutoff frequency of 2 kHz to remove low-frequency noise from the input signal.

2. To design and bias a BJT-based common-emitter amplifier that operates from a 9V DC supply and achieves a mid-band voltage gain of at least 60.

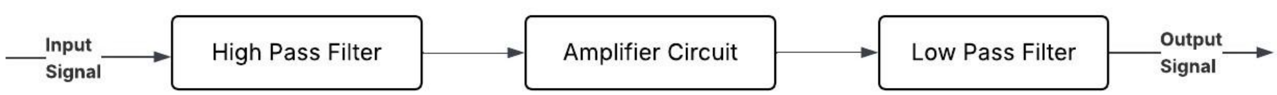
3. To design a first order low-pass RC filter with a cutoff frequency of 12 kHz to attenuate high frequency noise and smooth the output audio signal.

4. To select appropriate resistor and capacitor values through theoretical calculations and justify all component choices.

5. To measure and document input and output waveforms at each stage using an oscilloscope.

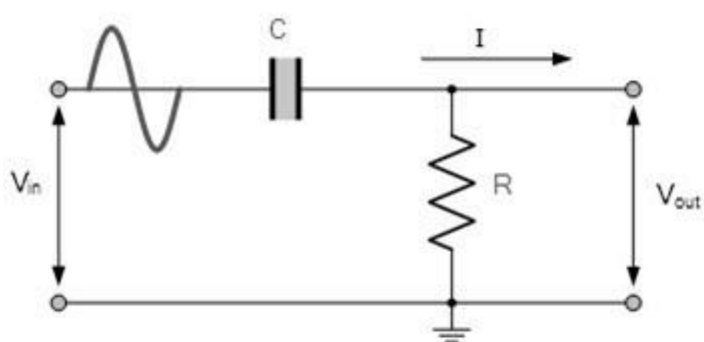
6. To evaluate the performance of the amplifier and the effectiveness of the filtering stages in improving signal quality.

Circuit Design Overview



Stage 1

High Pass Filter Design



Purpose of Using High-Pass Filters

In this AM audio output circuit, the main goal of the filter stage is to allow only the desired audio frequency band (2kHz – 12kHz) to pass through while rejecting unwanted low-frequency components (like hum, DC offset, and RF residue).

This design inserts the high-pass filtering function directly into the input coupling network of the common-emitter BJT amplifier, compared to a conventional method that uses external transmitted high-pass filters before to the amplifier stage. Improved signal integrity, fewer components, and lower stage-to-stage loading are some benefits of this integrated approach.

Circuit Configuration:
In this design, the high-pass filter is formed by the input coupling capacitor (C1) and the resister (R2).

HP_F: Capacitor Value = 10nF, Resistor Value = 8.2k ohms

Cutoff Frequency Calculation For high-pass RC Filter:

$$f_c = 1/2\pi RC$$

Substitute R = 8.2k Ω and C =10nF

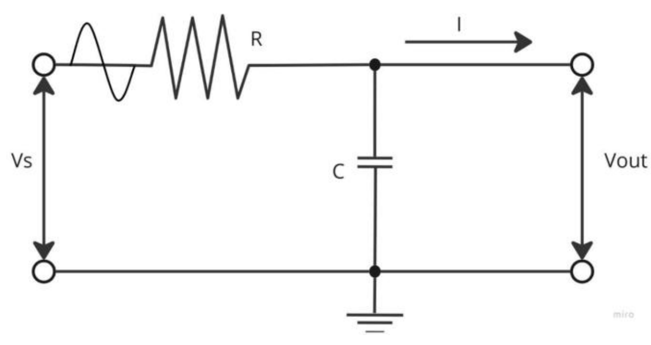
$f_c = 1 / 2\pi \times 8200 \times 10 \times 10^{-9}$

$f_c = 1.940\text{kHz}$

So, each HPF has a cutoff frequency of approximately 1.94 kHz.

Stage 2

Low pass Filter Design



Purpose of the LPF

- Suppress High-Frequency Noise: The LPF attenuates any remaining high-frequency components above 12 kHz, such as residual carrier or switching noise from the detector stage.

- Smooth Output Signal: It ensures the amplified signal delivered to the output (or indicator) is clean and free from unwanted high-frequency spikes.
- Complete the Band-Pass Response: Together with the HPFs, the LPF defines a band-pass region of approximately 2kHz –12 kHz, which covers the full audible range of interest for the AM receiver's audio output.

Cutoff Frequency Calculation

The cutoff frequency of this RC filter is given by

$f_c = 1 / 2\pi RC$

Substituting the component values:

$f_c = 1 / 2\pi \times 8.8 \times 10^3 \times 1.5 \times 10^{-9}$

fc= 12.05kHz

Hence, the LPF cutoff frequency is approximately 12 kHz, which matches the required upper limit of the audio band.

The output low-pass filter, designed with R = 8.8k Ω and C = 1.5n F, provides an upper cutoff around 12 kHz.

When combined with high-pass filter (1n F and 8.2k Ω, ≈ 2kHz cutoff), the overall circuit effectively functions as a band-pass filter passing the 2kHz – 12kHz range. This ensures only the desired audio frequencies are delivered to the amplifier output while suppressing both low-frequency hum and high-frequency noise — thus satisfying the design requirement.

Stage 3

BJT Amplifier Design

Amplification type & Purpose

The amplifier stage is a single-stage common-emitter (CE) BJT amplifier (Q1: 2N2222). The CE amplifier was chosen because it provides substantial voltage gain and a simple interface to the preceding filter stages. Its purpose in this circuit is to amplify the recovered audio (2–12 kHz) from the filter stage to a level suitable for an indicator or downstream stage while maintaining linearity and low distortion.

DC Biasing calculation (Q-Point)

Bias network and resistor values used in the amplifier:

Supply: V_{CC}=9 V

Voltage divider: R_1=47k Ω,R_2=8.2kΩ

Collector resistor: R_C=47k Ω

Emitter resistor: R_E=1000 Ω

Using the divider,

$V_B = Vcc \times R2/ (R1 + R2) = 9 \times 8.2k/ (8.2k + 47k) = 1.3369V$

$V_{BE} = 0.7v$

So, $V_E = V_B - V_{BE} = 1.3369 - 0.7 = 0.6369V$

Emitter DC current (approx):

$I_E = V_E / R_E = 0.6369 / 1000 = 0.63mA$

Collector current I_c = IE = 0.63mA (since $I_c = I_E$ for $\beta >>> 1$)

Collector Voltage:

$V_C = V_{CC} - I_C R_C = 9 - (0.63mA)(4700) = 6.039V$

So, the DC operating point (Q - point) is approximately:

$V_B \approx 1.336$ V

$V_E \approx 0.636V$

$I_E \approx 0.63$ mA

$I_C \approx 0.63$ mA

$V_C \approx 6.039$ V

$V_{CE} \approx 5.403$ V (in the active region → linear amplification)

Small – Signal parameters and gain calculation

intrinsic emitter resistance r′e

Small – signal intrinsic emitter resistance:

$r'e = 25mV / I_E$ [thermal voltage approximation at room temp is 25mV]

Using IE = 0.63mA

$r'e = 0.025 / 0.0063 = 3.968\Omega$

Approximate midband gain formula (frequency dependent)

The small-signal voltage gain magnitude (defined as the ratio of collector AC voltage to the small-signal base/emitter referenced input) is approximated by the following formula. Note that measured values are subject to clipping due to the high theoretical gain exceeding the power supply limits. $|A_v| = Vout / Vin$

Frequency	Vin	Vout	|Av|
5kHz	100mV	6.20V	62
12kHz	100mV	6.40V	64
15kHz	100mV	7.20V	72

Distortion, phase shift & clipping (discussion)

Distortion

Using a resistor at the emitter helps reduce distortion because it gives negative feedback.

A capacitor is added to partly bypass this resistor:

- At low frequencies (below 2 kHz), the capacitor does not work much → gain is low → distortion is low.
- At higher frequencies (above 2 kHz), the capacitor works more → gain becomes high (about 60+).

At high frequencies, if the input signal is too large, the transistor becomes non-linear and distortion increases.

To reduce distortion:

- Keep input signal small
- Use a smaller capacitor

Phase Shift

A common-emitter amplifier flips the signal. So, the output is about 180° opposite to the input. This is normal and does not affect sound quality in AM audio.

Other parts also change the phase depending on frequency:

- Input capacitor: causes a small phase change at low frequencies (around 2 kHz)
- Output capacitor: causes phase delay at low frequencies
- Transistor small capacitors: cause phase delay at high frequencies (above 12 kHz)

Overall:

- Phase shift is about 180° in the working range (2 kHz – 12 kHz)
- More small changes happen near low and high frequency limits

Clipping / Headroom

In this amplifier, the output signal size is limited by the DC voltage.

With a 9V supply, the circuit is set so:

- Collector voltage ≈ 4.5V (middle point)
- This allows the signal to swing up and down evenly

Typical values:

- Supply voltage = 9V
- Collector voltage ≈ 4.5V
- Emitter voltage ≈ 0.5 – 1V
- Maximum output swing ≈ 3.5 – 4V peak-to-peak

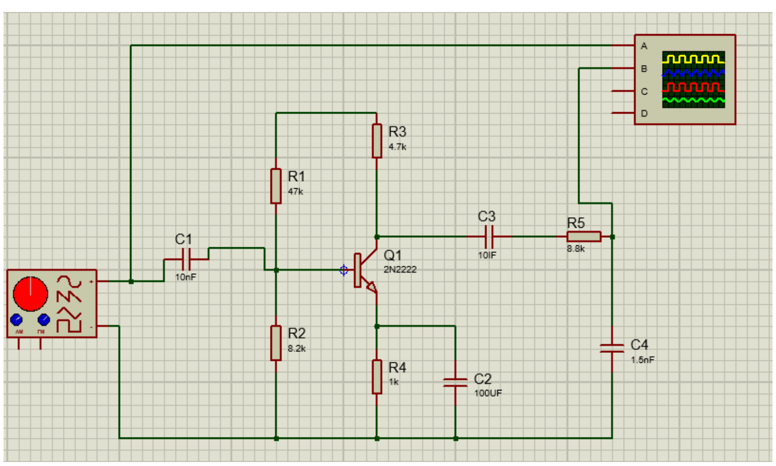
Clipping happens when:

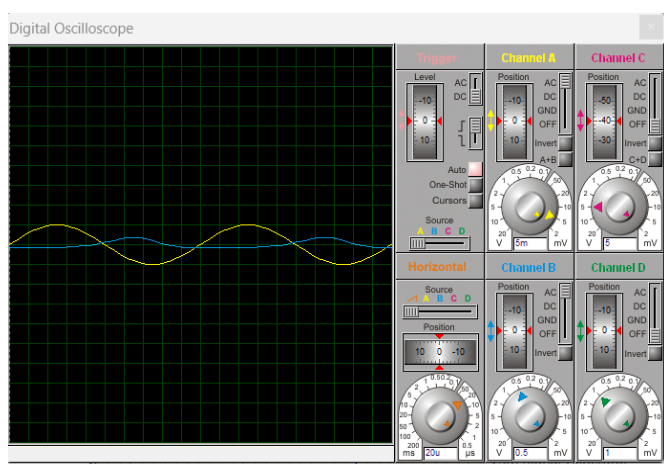
- Top clipping: Output tries to go above 9V → cannot go higher
- Bottom clipping: Transistor goes into saturation (very low voltage)

To avoid clipping:

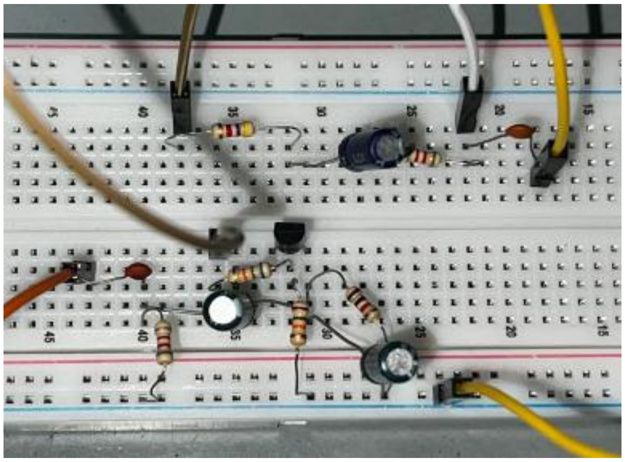
- Keep input signal small

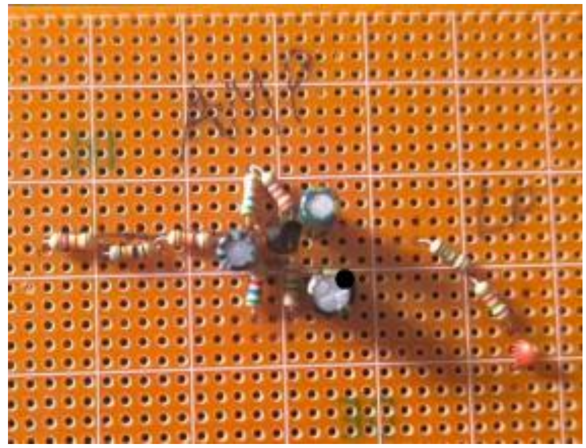
Circuit:





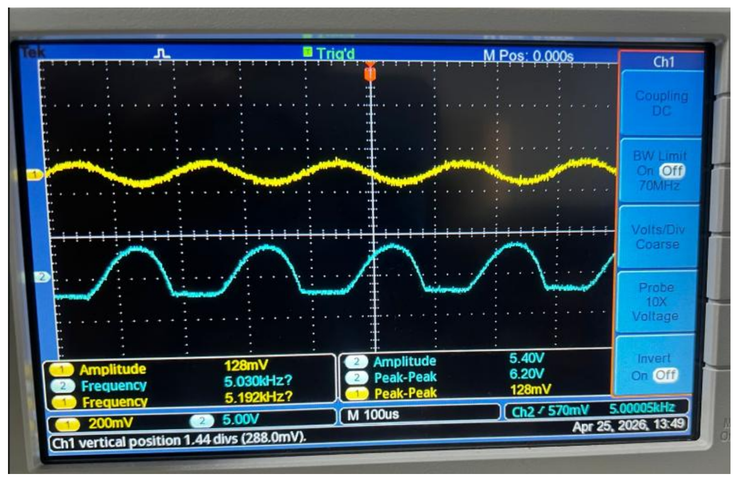
Proteus Circuit **Proteus Output**



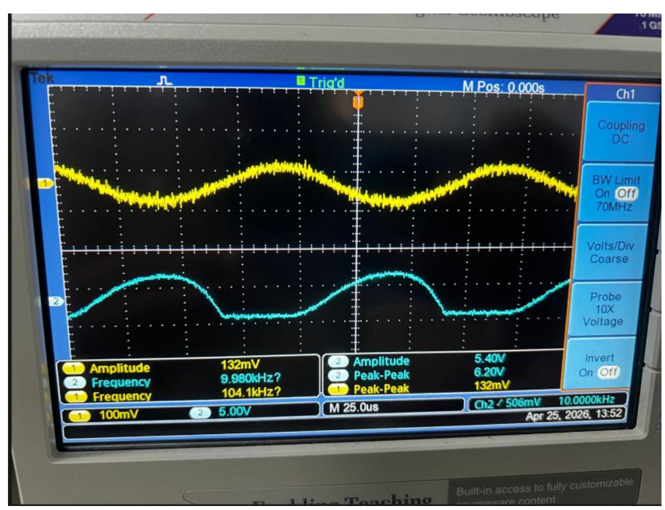


Breadboard Circuit **Dot board Circuit**

Some captures during lab times (Readings)

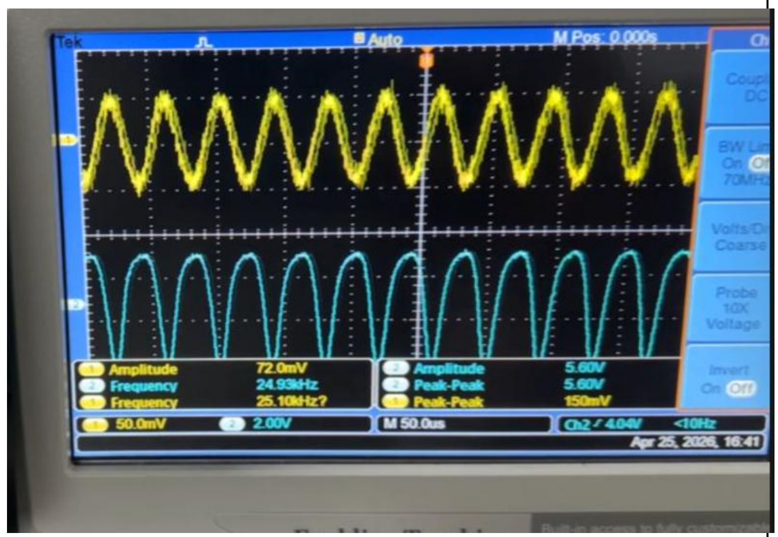


Set 100mV, 5kHz then we got 6.20V output that's mean we got 62 gain.



Set 100mV and 10kHz then we got 62V output that's mean we got 62 gains.

When the input frequency is set within this range(2kHz-12kHz), the amplifier operates properly and provides the expected high gain(60 or above 60).



Set 100mV and 25kHz then we got output 5.6V that's mean we got 56 gain

When the input is out of the range (below 2kHz or above 12kHz) we got the gain less than 60.

Discussion

The circuit works as a band-pass audio stage for an AM receiver using one high-pass filters, one low-pass filter, and a common-emitter amplifier.

It operates correctly in the 2 kHz to 12 kHz range, where the gain is 60 or above. Outside this range, the gain drops as expected due to the filters and circuit response.

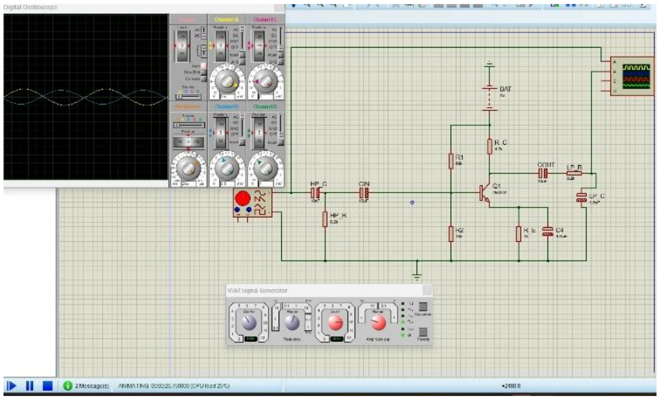
Overall, the design meets the required performance for the given frequency range.

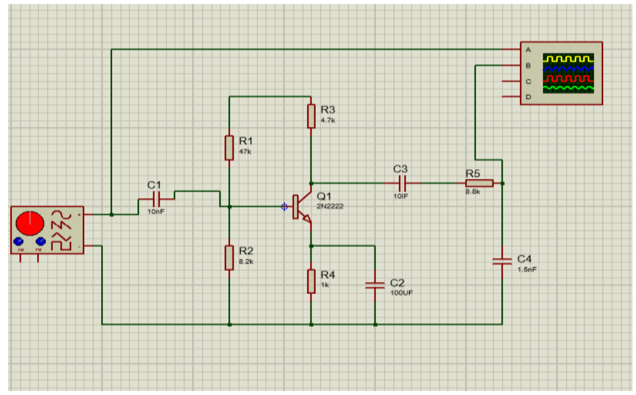
Gain and Amplifier Performance

Simulation results show that at 2 kHz, the gain is approximately 58–60, with slight variation. This satisfies the design requirement for the passband.

Within the operating range, the amplifier maintains high gain, and performance is stable enough for audio output use.

Problems Encountered and Solutions





3 Stage Circuit **2 Stage Circuit**

In my first circuit (the one with three stages), I planned to have a high-pass amplifier followed by a low-pass filter. However, even with three stages, the total voltage gain I measured was only around 35 to 40, which is less than my target of 60. This happened because each stage added only a small amount of gain, and the filter parts also reduced the signal. So, using three separate stages did not give me the high gain I needed.

To solve this problem, I tried a second circuit with only two stages. This circuit does not use a separate high-pass filter block. Instead, I combined the high-pass filtering with the amplification by carefully choosing the capacitors and resistors around the transistor. With this two-stage design, I successfully got a gain of 60 or more. The reason it works better is that fewer stages mean less signal loss, and the filtering is built directly into the amplifier instead of being separate.

Conclusion

Within the intended frequency range of 2 kHz to 12 kHz, the band-pass amplifier circuit's design was successful in achieving a gain greater than 60 times. The output signal's stability and quality were greatly enhanced by the addition of the op-amp buffer. All things considered, the design successfully filtered and amplified the target frequency band.

References

1. Lecture Notes

2. Data sheets (BJT, Diodes)